Response to Reviewers

PLOS ONE, PONE-D-26-04349 (Osgood-Zimmerman & Raredon)
Originally submitted as "Spatial Modeling of Tissues for Morphogenic Field Analysis"

Draft of 2026-10-23 | Revision deadline: December 2026

INTERNAL DRAFT, NOT FOR SUBMISSION.

One vocabulary throughout, three states. [DONE] finished and verified, nothing remains; [INCOMPLETE] started, with real work already in the manuscript, but not finished; [TODO] not started.

[. . .] labels a reviewer item as a whole. The red inline markers **[INCOMPLETE: . . .]** and **[TODO: . . .]** mark one specific outstanding task within an item, and always carry the same word as that item's status, so the two can never disagree. A [DONE] item carries no marker.

Every outstanding task also appears in the consolidated Open items section below. Before submitting, set `\draftmodefalse` and confirm no marker remains.

Dear Dr. D'Acquisto and Reviewers,

Thank you for the careful and constructive reviews. We are grateful for the time invested, and we think the manuscript is substantially better for the revision.

We should say at the outset that the revision goes further than point-by-point repair. Working through the reviewers' comments on positioning (R1.1, R2.10), on controls and baselines (R2.8) and on the reach of our claims (R2.11) led us to a clearer statement of what the framework offers that existing methods do not. What it offers is a posterior distribution over every gene field and, from it, over every ligand-receptor interaction field. The revised manuscript makes the use of that posterior its central result: it asks whether differences that are visible on maps of the interaction fields are supported by the data, for averages over regions, along transects and over whole fields, and it sets each answer beside the readings commonly made without a posterior. To do this we refitted every gene with the published model to obtain joint posterior draws, which the original analysis had not stored. The regions, transects and examples were fixed before any posterior comparison was computed. On the section studied, many differences that look clear on the maps are not supported once the uncertainty, and in particular its spatial correlation, is accounted for. The analyses that the original submission centered on, the tissue domains and the controls that the reviewers asked for, are retained in a shorter section, with their full text in a supporting appendix.

Reviewer 2 asked whether the framework identifies biologically meaningful spatial structure beyond smoothing, increased feature dimensionality, or clustering. We have addressed that question directly with three controls (Section R2.8): a coordinate shuffle with full model refitting, which collapses the domains; a concordance test against the independent anatomical annotation published with the source dataset; and the ligand-receptor pair-shuffling control the reviewer suggested, which found no effect of the pairing on domain coherence and which we report in full, together with the limit it places on our conclusions. The reframed analysis also answers the reviewer's request for a baseline: every comparison is now set against the plug-in reading, a test that treats locations as independent, and the posterior with its spatial correlation removed.

We have also moderated the generalizability and biological-interpretation language throughout, clarified the statistical model and its implementation, and added a supplementary table mapping every mathematical object in the manuscript to the code that implements it.

Reviewer comments are reproduced verbatim in italics, followed by our response. Because the changes are extensive, a word-level comparison is hard to read; we suggest reading the clean revised manuscript together with the revision summary that follows this letter, which lists what moved, what is new and what was removed. The required marked-up manuscript is also provided. One caveat about that file: the markup covers the prose but not the interiors of displayed equations, so the revised model specification in the Materials and methods does not appear visually flagged even though it changed. We describe that change in words under R2.1 below.

Sincerely,
Aaron Osgood-Zimmerman and Micha Sam Brickman Raredon

Summary of the revision

This summary is meant to be read with the clean revised manuscript. Figure and section references are to the revised manuscript unless stated otherwise.

Title, abstract and Introduction. The title and abstract are rewritten around the central question. The Introduction keeps its three opening subsections, shortened, and adds a final subsection, "Reading interaction fields under uncertainty", which states the question, the related work, and what the paper does.

Materials and methods. The model specification is as revised for R2.1 to R2.4. A new subsection, "Uncertainty-aware comparison of interaction fields", describes: the joint posterior draws obtained by refitting every gene with the published specification, and the check that the refits reproduce the published fits (778 of 790 genes within Monte Carlo error; the 12 that do not, all sparsely detected, enter 17 of 1,477 pairs, which are flagged and reported both ways); the posterior of an interaction field; the fixed boxes, published anatomical regions and transects, chosen before any comparison; the quantities compared; the joint inference (excursion sets and simultaneous bands); the three readings we compare against; and how the examples were selected.

Results. Reordered. After the model comparison (Figs 3 and 4, Table 1, essentially unchanged), a new comparison of the plug-in interaction fields with their posteriors (Fig 5) motivates the new sections: region comparisons (new Fig 6 and Table 2), transects (new Fig 7) and whole fields (new Fig 8). The tissue-domain analysis follows in a shorter section (Fig 9, formerly Fig 6), with the three controls condensed to one paragraph.

Moved to Supporting Information. The comparison of convolution operators (formerly Fig 5, now S15 Fig) and the full text of the operator comparison and of the three controls (new S16 Appendix, moved verbatim apart from cross-references).

New Supporting Information. S11 Fig (the fixed regions and transects, and the five examples on which the "looks different" threshold was set), S12 Fig (all example region comparisons), S13

Table (reproduction of the published fits, and every count with and without the flagged pairs), S14 File (per-pair results).

Discussion. New opening paragraphs on what the uncertainty adds, why treating locations as independent fails here, and the limits of the joint statements. "Relationship to existing methods" adds the closest precedents for each part of the new analysis and restates our contribution more narrowly. The paragraph that said uncertainty was not propagated is replaced: it is now propagated for the comparisons, but not through the clustering.

Removed. The claim, in the original Introduction, that the results show domain architecture can be derived from ligand-receptor patterns alone, which the pair-shuffling control does not support in that form.

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1. Open items: everything still outstanding

Internal only. This section disappears with `\draftmodefalse`**.** Rewritten 2026-10-08; the July list it replaces is in git history. The plain-language account of the whole revision is `REVISION-SUMMARY-2026-10-08.org`.

A. Needs Sam

1. **Title**, to be agreed at the end (placeholder in `paper-plos.tex`).
2. **Graphical abstract (R1.5)**: the 2026-10-08 draft shows the domains, which are no longer the central result; to be redone for the uncertainty message.
3. **Fig 1** redraw to include the uncertainty step; **Figs 1 and 2** text below 8 pt at print size (rough pixel measurement 2026-10-23).
4. **Biology read** of the example comparisons.

B. Author sign-off on substantive text

5. The reframing as a whole (letter opening, revision summary, R1.1, R2.8, R2.10).
6. New Methods subsection, new Results sections, Discussion opening, contribution statement.

C. Mechanics, immediately before upload

7. Set `\draftmodefalse` here and confirm `python3 check-response-letter.py` reports no markers.
8. In `paper-plos.tex` set `\embedfigsfalse` and upload `submission-figures/Fig1.tif` to `Fig9.tif` (`bash export-submission-figures.sh`).
9. Upload S9 Table, S10 to S12 Fig, S13 Table, S14 File, S15 Fig, S16 Appendix (`supporting-information/`).
10. Build the manuscript, the required marked-up version (`latexdiff --math-markup=off` against tag `rev1-baseline`, uploaded as "Revised Article with Changes Highlighted") and this letter, following `README-build.md`.
11. Finalize the cover letter (`cover-letter-draft.org`).

2. Status summary

Internal only, delete this section before submission.

Item	Ask	Status
JR-1	PLOS style and file naming	Done
JR-2	Role of funder statement	Done
JR-3	Remove funding text from Acknowledgments	Done
JR-4	PLOS LaTeX template	Done
JR-5	Upload Supplemental Materials as SI	Done
JR-6	Evaluate recommended citations	Done
R1.1	Comparative analysis vs existing methods	Done
R1.2	Tissue sample and methods detail	Done
R1.3	Figures too small, reorganize	Done
R1.4	Small-scale disease proof of concept	Done
R1.5	Better graphical abstract	Incomplete, to be redone for the new message
R2.1	Define model parameters and priors at first use	Done
R2.2	Notation-to-code supplementary table	Done
R2.3	Clarify that final analyses used observed nUMI	Done
R2.4	Define metrics; Table 1 abbreviations and percentages	Done
R2.5	Rationale for Cd44 as representative feature	Done
R2.6	Elongated-pixel distortion in Figs 3 and 4	Done
R2.7	Clearer captions for Figs 5 and 6	Done
R2.8	**Controls and baselines**	**Done, three reported**
R2.9	Reproducibility documentation	Done
R2.10	Positioning vs related methods	Done
R2.11	Moderate generalizability claims	Done
R2.12	Cautious biological language	Done

3. Journal requirements

JR-1. PLOS ONE style requirements and file naming

[DONE]

> *Please ensure that your manuscript meets PLOS ONE's style requirements, including those for file naming.*

The manuscript has been reformatted to the PLOS ONE style templates. Section headings, title capitalization, the reference style, and file naming now follow the PLOS ONE guidelines. See also JR-4.

JR-2. Role of the funder

[DONE]

> *Please state what role the funders took in the study. If the funders had no role, please state: "The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript."*

We confirm the following statement, which we have included in our cover letter for the journal to enter into the online submission form:

> The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

JR-3. Funding text in the Acknowledgments

[DONE]

> *We note that you have provided funding information that is not currently declared in your Funding Statement. However, funding information should not appear in the Acknowledgments section or other areas of your manuscript.*

The Acknowledgments section now contains only non-financial acknowledgments, and we have checked the full manuscript text for funding, grant, and sponsor language and confirmed that none remains anywhere in it. The funding information appears solely in the Funding Statement, and our updated statement is reproduced in the cover letter.

JR-4. PLOS LaTeX template

[DONE]

> *Please update your submission to use the PLOS LaTeX template.*

The manuscript has been ported from the Elsevier `elsarticle` class to the PLOS LaTeX template, including the PLOS preamble, title and author block, and the `plos2025` bibliography style.

JR-5. Supporting Information upload

[DONE]

> *Please upload a copy of Supporting Information Figure/Table/etc. "Supplemental Materials" which you refer to in your text on page 19.*

We have added genuine Supporting Information files, all uploaded with this revision. S9 Table gives the notation-to-code mapping requested in R2.2, and S10 Fig summarizes the controls requested in R2.8. The reframed analysis adds S11 Fig (the fixed regions and transects, and the examples on which our threshold for a visible difference was set), S12 Fig (all example region comparisons), S13 Table (reproduction of the published fits by our refits), S14 File (per-pair results, about 7 MB), S15 Fig (the comparison of convolution operators, formerly Fig 5) and S16 Appendix (the operator comparison and the controls in full). We have also confirmed that every S-number cited in the text now matches its intended item; S1 and S9 had no citation in the original text, and the S3 through S8 pointers were reassigned, because the original manuscript cited "Supplement 3/4/5" under a numbering that did not correspond to the S1–S8 list.

We must, however, be straightforward about the rest. The eight supplements are research data archives rather than supplementary documents: together they are 53.5 GB across 12,275 files, and the largest single file is 11.6 GB. They cannot be uploaded as Supporting Information files, and we do

not think a reader would want them to be. They remain deposited externally and are cited individually at the appropriate points in the text. We have prepared a manifest recording the exact size, file count, and content type of each, which is in the repository at `Submission_PLOSOne/submission-2/supporting-in`

We would welcome the editor's guidance on the preferred arrangement. We undertake to convert these deposits from private share links to public, DOI-minted records on acceptance, so that the published paper cites permanent identifiers. We have deliberately not minted them yet, because the deposited contents may still change in review and a DOI should point at a final record rather than at one we expect to revise. We can also prepare small curated extracts of individual supplements if the journal would prefer a reader to be able to open something without a large download.

JR-6. Recommended citations

[DONE]

> *If the reviewer comments include a recommendation to cite specific previously published works, please review and evaluate these publications to determine whether they are relevant and should be cited.*

We evaluated every method named by Reviewer 2 (SpatialDE, BOOST-GP, Spatial Variance Component Analysis, and SpatialDM) and judged all of them relevant. Each is now cited and discussed in a new positioning subsection; see our response to R2.10. We have additionally cited CytoSignal (Liu et al., *Nature Genetics*, 2026), concurrent work that was not available at the time of our original submission and that analyzes the same source dataset, and the work on uncertainty and spatial correlation that the reframed analysis rests on or is closest to (see R2.10).

4. Academic editor

> *It seems to me that the Reviewers have asked a number of clarifications and more detailed explanation for the approaches used in the study while finding the overall idea of interest. I would suggest the authors to fully address the comments made by the Reviewers as this might significantly improve the impact of the study and its overall take-home message.*

We agree, and we have treated every reviewer comment as actionable. The clarifications requested by Reviewer 2 concerning the statistical model, its implementation, and the evaluation metrics were fair; the manuscript understated how much detail a reader needs to reproduce or critically assess the work. The point-by-point responses below describe each change.

5. Reviewer 1

R1.1. Comparative analysis with existing spatial modeling techniques

[DONE]

> *Recommend the authors to perform a comparative analysis with other existing techniques for spatial modelling techniques as verifications.*

We have added comparison and validation along four axes. The first is new to this revision and is now the central result; the other three are described in full in our response to Reviewer 2's comment R2.8.

First, every comparison the revised manuscript makes between regions, along transects or over whole fields is set beside the ways the same question is commonly answered without our posterior:

reading the point-estimate map, a Wilcoxon test that treats each location as an independent observation, and our own posterior with the correlation between locations removed. Across 19,201 comparisons between pairs of regions (1,477 ligand-receptor pairs, 13 fixed region comparisons each), the Wilcoxon test finds a difference in 90%, the posterior with its spatial correlation removed in 73%, and the joint posterior in 46% (Table 2). Of the comparisons that look different on the maps by a factor of at least 1.25, 31% are not supported by the posterior. These counts come from one tissue section and share regions and genes, so they are not rates, but they show concretely where the framework changes a conclusion that a simpler reading would draw.

Second, within our own framework we compare candidate convolution operators quantitatively, and this comparison includes the product operator used by CytoSignal. On this dataset, and in a single clustering run, the common-minimum operator we adopt gave more clusters and more spatially coherent domains than the product operator. This comparison is now S15 Fig.

Third, we compare the emergent domains against an external reference that we did not produce: the anatomical annotations published with the source dataset by Chen et al. (2022). This is a comparison to an independently derived, expression-based segmentation of the same tissue section.

Fourth, we added a spatial permutation test with full model refitting, which is the standard validation device in this literature.

We want to be straightforward about what we have not done. We have not run a full method-versus-method benchmark against SpatialDE, BOOST-GP, SVCA, SpatialDM, or CytoSignal. Those methods answer different questions from ours: they test individual features or interactions for spatial significance, whereas our contribution is a posterior over continuous fields and its use to judge whether apparent differences are supported. There is no shared target quantity on which a head-to-head accuracy comparison would be meaningful without first defining a benchmark task, which we regard as its own piece of work. We have instead positioned our approach against each of these methods, and against the closest precedents for the new analysis, explicitly (R2.10), and we state this limitation in the Discussion rather than leaving it implicit.

R1.2. Detail on the tissue sample and methods

[DONE]

> *The methods on the tissue/patient sample obtained here is missing and lacks details.*

We have expanded the Dataset selection and Dataset preprocessing subsections. We should first clarify one point that the comment may turn on: there is no patient sample here. The material is a mouse embryo section taken from a public atlas, we collected no new tissue for this study, and we did not perform the tissue processing or sequencing ourselves, so the experimental protocol is properly the source publication's to report. We now say this explicitly rather than leaving it to be inferred.

The revised text states the source publication and resource (Chen et al. 2022, the Mouse Organogenesis Spatiotemporal Transcriptomic Atlas), the platform and the nature of its capture chemistry, the developmental stage and section identity (embryonic day 16.5, section E16.5_E1S3), the field of view analyzed (x from 15000 to 25000 and y from 11500 to 18000 in raw chip coordinates), the binning applied (bin 50, giving a 200×130 grid of 26,000 spatial locations), and the filtering that produced the analyzed feature set, including the attrition from 918 curated features to the 912 with converged fits and the 1477 ligand-receptor pairs computable from them. We have also removed

the sentence describing the dataset as "beautiful," which was not a reason for selecting it, and replaced it with our actual reasons.

R1.3. Figure size and organization

[DONE]

> *The data presented in the figure is on the small side of the spectrum, recommend the authors to reorganise them.*

We have rebuilt Figures 3 to 6 of the original submission from scripts at the full PLOS page width, with all text set at 8 point or larger at printed size; in the original submission some labels were drawn large and then reduced, so that they printed at about 3 to 4 point. Figures 3 and 4 have also been corrected for the aspect-ratio distortion noted by Reviewer 2 (R2.6), and the original Figures 5 and 6 for the same defect, which the reviewer did not flag but which was present there too. The four new figures of the revised manuscript (Figs 5 to 8) were built at their printed size with the same standard.

Figure numbers have changed with the reframing. The original Figure 5, the comparison of convolution operators, is now S15 Fig, and the original Figure 6, the tissue domains, is Fig 9.

For Figures 3 and 4 we kept the panel arrangement; their difficulty was size rather than ordering. The original Figures 5 and 6 we reorganized, because at the page-height limit their original arrangement could not be made legible. The operator comparison (now S15 Fig) shows each pair's ligand and receptor fields once, in a row of their own, followed by one row per convolution operator, so the operators can be compared by reading down a column. The domain figure (now Fig 9) places the five feature depths in columns rather than rows, and replaces the five repeated summary panels with a single row of summary curves that marks the five depths shown; that row also carries the random-subset control described under "Additional corrections".

R1.4. Small-scale disease proof of concept

[DONE]

> *Since the author claims the spatial modelling provides multidimensional information to guide disease assessment, it might be good for the author to use this paper to perform a small scale POC work.*

We appreciate this suggestion and have thought about it carefully, but we have concluded that adding a disease dataset is not the right change for this manuscript, and we have instead removed the framing that invited it.

The manuscript did not report a disease result; disease was raised only as future motivation. On reflection, that framing promised more than the work delivers, which is also the substance of Reviewer 2's comment R2.11 about generalizability. Rather than adding a second dataset, we have moderated the disease language so the stated scope matches the evidence: this is a methodological proof of concept on healthy developing tissue.

We think this is the more honest resolution. A disease proof of concept done properly requires matched healthy and diseased tissue, attention to the confounding between disease state and tissue composition, and enough replication to distinguish a disease effect from between-sample variation. Done in the time available on a single additional section, it would produce an illustrative figure that we could not defend statistically, which is the opposite of what this manuscript argues for. We

have noted the application to diseased tissue as clearly future work in the Discussion, alongside the lung-tissue direction already mentioned there.

R1.5. Graphical abstract

[INCOMPLETE]

> *Minor comment: The graphical abstract submitted did not do the work justice, consider a better images rather than copy-pasting from the data section.*

We agree, and we will replace the graphical abstract with one designed for the purpose rather than copied from the figures, drawn directly from the data and showing the central result of the revised manuscript.

[INCOMPLETE: The 2026-10-08 draft (`figures-regenerated/graphical-abstract.png`) shows the tissue domains, which are no longer the central result; to be redone for the uncertainty message and reviewed by Raredon before submission.]

6. Reviewer 2

We thank Reviewer 2 for an unusually thorough and useful review. The comments identified real gaps, particularly the absence of controls, and the manuscript is materially stronger as a result.

R2.1. Precision of the statistical model specification

[DONE]

> *First, the statistical model is not specified with sufficient precision. The manuscript introduces a hierarchical statistical framework, but several key parameters are not clearly defined, including θ_i, θ_N, $\Sigma(\theta_i)$, $\Sigma(\theta_N)$, the covariance function, prior structure, and the relationship between the abstract mathematical model and the computational implementation. Since this model is central to the manuscript, all parameters should be introduced clearly at first use.*

We accept this criticism. The model section was written at a level of abstraction that made the framework readable but left the actual fitted model underdetermined. We have rewritten it to define every symbol at first use and to state the concrete choices used in the reported analyses.

Specifically, the revised Materials and methods now state that the spatial random effect for feature i is a zero-mean Gaussian process with a Matérn covariance function, represented through the stochastic partial differential equation approach, so that $\theta_i = (\rho_i, \sigma_i)$ collects the range and marginal standard deviation for feature i, and $\Sigma(\theta_i)$ denotes the resulting Matérn covariance matrix evaluated at the mesh nodes. The smoothness parameter is fixed rather than estimated ($\alpha = 2$, giving Matérn smoothness $\nu = 1$ in two dimensions). We place penalized-complexity priors on θ_i: the range prior satisfies $P(\rho_i < 500) = 0.95$ and the marginal standard deviation prior satisfies $P(\sigma_i > 0.1) = 0.05$, with the range in the same physical units as the chip coordinates. A sum-to-zero constraint is applied to the field for identifiability against the intercept. The symbols θ_N and $\Sigma(\theta_N)$ played a role only in the latent total-count model, which the final analyses did not use; see R2.3, where we have clarified their status rather than leaving them ambiguous.

We have also made two changes to the model display itself. The process stage is now written additively, as $\log \lambda_i(\mathbf{s}) = \alpha_i + u_i(\mathbf{s})$ with u_i a zero-mean Gaussian process, rather than as a Gaussian process with mean α_i. The two are equivalent, but the additive form separates the intercept from the field, which is how the model is actually implemented, and it gives the field its own symbol so that

it can be named in the mapping table. We have also corrected a small imprecision: the original text wrote $F_i(\mathbf{s}) = \mathrm{N}(\mathbf{s}) \times \lambda_i(\mathbf{s})$, equating a random variable with its mean, where $\mathrm{E}(F_i(\mathbf{s})) = \mathrm{N}(\mathbf{s}) \times \lambda_i(\mathbf{s})$ was intended.

We have added an explicit paragraph connecting the abstract model to its implementation, including the SPDE representation, the mesh, the inference method, and the number of posterior draws, and a supplementary table making the mapping object by object (R2.2).

R2.2. Notation-to-code mapping

[DONE]

> *Second, the correspondence between the notation in the manuscript and the public code repository is difficult to follow. Readers should be able to identify where mathematical objects such as $\lambda_i(s)$, $F_i(s)$, $N(s)$, α_i, $\Sigma(\theta_i)$, and θ_i are implemented in the code. I recommend adding a supplementary table mapping each manuscript term to its corresponding code object, script, or model component.*

We have added this table as Supporting Information (S9 Table). Each row gives a mathematical symbol, its meaning, the name of the corresponding code object, and the script and line at which it is constructed, so that a reader can move from any equation in the manuscript to the lines that implement it. It covers every symbol the reviewer names, plus the mesh, the Matérn SPDE object, the priors, the likelihood, the posterior draws, and the four prediction scores. Every path, object name, and line number in it was checked against the files rather than written from memory.

Two things in the table are worth flagging here, because both are traps for a reader navigating the repository. First, the file-level default prior at line 32 of the model script sets the marginal standard deviation to 1, but it is overridden inside the Poisson branch at line 63, which sets it to 0.1; the manuscript quotes the line-63 values, because the Poisson branch produced every reported result. Second, the absolute error uses the posterior predictive *median* while the squared error and Dawid-Sebastiani score use the mean. The manuscript equations now reflect that difference, which the original text did not.

R2.3. Treatment of total counts $N(s)$

[DONE]

> *Third, the treatment of total counts $N(s)$ requires clarification. The manuscript presents a latent model for $N(s)$, but later states that modeling $N(s)$ did not substantially alter the results and that observed nUMI values were used directly. The authors should clearly state whether the final reported analyses used a joint latent model, an observed offset or exposure term, or a simplified model using observed nUMI values.*

The reviewer is right that this was ambiguous, and we have stated it plainly.

All analyses reported in the figures and in Table 1 use the observed number of unique molecular identifiers at each location as a fixed offset, that is, as an exposure term entering the Poisson mean multiplicatively and treated as known rather than modeled. No result in the manuscript comes from the joint latent model for $N(s)$.

We retain the latent formulation in the model development section because it is the general form of the framework and because we tested it, but we now flag its status explicitly at the point of first presentation, in a short subsection immediately following the model display, rather than only in passing in the Results. That subsection states which lines of the model specification the reported analyses actually use, and that every symbol carrying an N subscript plays no part in any reported result.

On the comparison itself, the original manuscript said only that modeling $N(s)$ "didn't practically alter the outputs," which was an impression rather than a measurement. We have now measured it. Twenty-three features were fitted under both specifications, and comparing the fitted feature fields point by point across all 26,000 locations gives a Pearson correlation between the joint and offset fits of 0.986 to 0.997, with a median of 0.995. The levels differ more than that similarity in pattern suggests: the mean absolute difference between the two fitted fields, as a percentage of the mean fitted count for that feature, ranges from 5.1% to 13.5% with a median of 6.6%.

We think this supports the choice we made while also being more honest than the original wording. The spatial pattern, which is what every downstream step in this manuscript uses, is essentially unchanged; the absolute level is not identical, and we no longer imply that it is. The revised text gives these figures and states the conclusion in that restricted form: we adopted the offset specification because it preserves the structure the analyses depend on and is far cheaper to fit, not because the two specifications are interchangeable for any purpose.

R2.4. Prediction metrics and model comparison

[DONE]

> *Fourth, the prediction metrics and model comparison need clearer explanation. The manuscript compares Poisson, ZIP, and ZAP models using MAE, RMSE, Dawid-Sebastiani score, and log score, but the definitions, orientation, and interpretation of these metrics are not sufficiently clear. Table 1 should define all abbreviations and explain the percentages in parentheses. The authors should also clarify which criteria were prioritized and why the evidence supports selecting the Poisson model over ZIP and ZAP.*

We have added explicit definitions for each metric, stated the orientation of each, expanded the Table 1 caption to define every abbreviation and the parenthetical percentages, and stated which criteria we prioritized and why.

In summary: mean absolute error (MAE) and root mean squared error (RMSE) are point-forecast accuracy measures; the Dawid-Sebastiani score (DS) and logarithmic score (LS) are proper scoring rules that assess the full predictive distribution rather than a point summary. All four are negatively oriented, so smaller values indicate better performance. We prioritized the proper scoring rules, because the manuscript's argument concerns the quality of the predictive distribution and its uncertainty rather than point accuracy alone.

We have also rewritten the Table 1 caption completely, and in doing so we found and corrected an error in how the table had been read.

The parenthetical percentages are the mean across features of the ratio of that model's score to the Poisson score *for the same feature*, so 100% denotes a model indistinguishable from Poisson. They are ratios, not differences, and they are not the ratio of the two averages printed in the table, which is why 0.29 against 0.21 appears as 111% rather than 138%. The old caption called them an "average percentage difference," which is a misnomer for a ratio; the new caption defines them correctly and defines every abbreviation.

The error concerns the Dawid-Sebastiani column, reported in the table as MDS, the mean Dawid-Sebastiani score. Because MDS is negative, a ratio below 100% corresponds to a score *nearer zero*, and DS is negatively oriented, so a ratio below 100% indicates worse performance rather than better. Reading the 85% and 82% entries as ZIP and ZAP outperforming Poisson, which is the natural reading and one we had made ourselves while preparing this response, inverts the result. Comparing the raw values directly, Poisson at -2.15 is better than ZIP at -1.78 and ZAP at -1.77.

Poisson is therefore the best model on all four scores, not three of four, and no criterion disagrees. The new caption warns the reader about this explicitly and directs them to the raw MDS values.

On which criteria we prioritized: we prioritized the two proper scoring rules over the point-forecast measures, because the manuscript's argument concerns the quality of the predictive distribution rather than point accuracy. As it happens the choice does not matter here, since Poisson wins on all four.

We have added one piece of new evidence, because the averages alone do not show whether the result is driven by a few extreme features. Comparing the three likelihoods feature by feature, Poisson gives the lowest score for 814 of the 912 features on MAE, 880 on RMSE, 734 on MDS, and 829 on MLG, and is best on all four simultaneously for 645 of the 912. These counts are computed from the same per-feature score files that produce Table 1.

Finally, the caption's claim that averages were taken over "all 918 modeled ligand/receptor features" was wrong in three ways, and is now corrected. The fitted set is 912, not 918. The averages exclude feature-by-model combinations with extreme values indicating nonconvergence, via the filter `MAE < 100 & abs(MDS) < 100`. And that filter is applied to the absolute scores for one table and to the *ratios* for the other, so the two halves of the table exclude different numbers of rows: 4 of the 2,736 combinations for the absolute averages and 7 for the parenthetical ratios. In both cases only the ZIP row is affected, so ZIP is averaged over 908 features for the absolute values and 905 for the ratios, while Poisson and ZAP use all 912.

R2.5. Rationale for Cd44 as the representative feature

[DONE]

> *For example, the rationale for selecting Cd44 as a representative example should be explained.*

We have added the rationale to the text and to the figure caption, and the honest answer is that Cd44 was chosen for clarity of illustration rather than for any property that makes it special.

The point Fig 3 makes is that count-normalization can change spatial topology. Many features show that effect. Cd44 shows it against an unambiguous anatomical landmark, the developing spine, which makes the raw and normalized panels easy to compare side by side in adjacent plots. That is why it was picked. We now say so, rather than leaving a reader to infer that Cd44 was somehow representative in a statistical sense.

Because a single chosen example invites exactly the reviewer's question, we also point the reader to where the same comparison can be checked more broadly: S3 File gives it for a further 23 selected features, and S2 File gives per-feature outputs for the whole curated set. A reader who suspects we cherry-picked can therefore verify how general the pattern is, which we think is a better answer than asserting that we did not.

R2.6. Elongated-pixel distortion in Figures 3 and 4

[DONE]

> *Figures 3 and 4 appear visually distorted due to elongated rectangular pixels, which should be corrected or clearly identified as a visualization artifact.*

The reviewer is right, and we are grateful for the catch, because the cause turned out to be a genuine bug rather than a display preference.

We first checked whether the bins are actually square, since if they were not, the honest response would have been to say so rather than to rescale. They are: the binning routine steps the x and y breaks by the same `agg.size`, and the Stereo-seq spot pitch is isotropic, so a bin is 50 by 50 capture spots in both directions.

The distortion came from the plotting call. Every panel was drawn with `asp = qp.res.y / qp.res.x`, which for this field of view is $130/200 = 0.65$. In R base graphics `asp` is the ratio of a y-unit length to an x-unit length, so setting it to 0.65 compresses the y axis relative to x and stretches every bin horizontally by $1/0.65 = 1.54$. The correct value is `asp = 1`. It appears the aspect ratio of the *domain* was supplied where the aspect ratio of the *units* was required.

We have corrected this at source rather than only in the affected figures: all 151 occurrences across the pipeline scripts now use `asp = 1`, so any figure regenerated from this point is correct. Figures 3 and 4 have been regenerated from a script at the correct aspect ratio. The original Figures 5 and 6 (now S15 Fig and Fig 9) had a separate, unflagged version of the same defect: their spatial panels were drawn by a plotting routine that stretches each panel to fill the space given to it. We have corrected those as well, fixing the aspect ratio to the true 200 by 130 bin geometry, and rebuilt both figures from scripts. All four scripts are in the repository at `Submission_PLOSOne/submission-2/figures-regenerated/`.

R2.7. Captions for Figures 5 and 6

[DONE]

> *Figures 5 and 6 require clearer captions, definitions of abbreviations, explanation of rows and columns, and a more explicit connection between the visual evidence and the conclusions drawn.*

With the reframing, the original Figure 5 is now S15 Fig and the original Figure 6 is Fig 9; the captions travel with them. We have rewritten both captions to be self-contained: each now defines every abbreviation used in the panel, states what the rows and columns represent, and says explicitly what the reader should take from the figure and how it supports the claim made in the text.

Concretely, the Figure 5 caption now identifies the three convolution operators (with the code labels `m.est.prod`, `m.est.gm` and `m.est.min` used in the code), the four example ligand-receptor pairs and their input fields, the interaction field each operator produces, the resulting tissue domains and the corresponding entrogram, and what the two summary panels plot. It states that cluster colors are arbitrary labels, so that the comparable quantities across columns are the number of domains and their spatial organization, and it says which direction of the summary scatter is favorable.

The Figure 6 caption now states the five feature depths, identifies each row, explains the entrogram scale and how to read a coherent domain from it, explains the summary curves (including the random-subset control), and identifies the eight three-dimensional surfaces and the sub-region they cover. It also carries caveats that the figure cannot show on its own: that the fields are model-derived estimates rather than measurements of binding, and that each domain map is a single run of a clustering step whose exact reproducibility we have not established.

Both figures were also reorganized for legibility; see our response to R1.3.

R2.8. Controls and baselines

[DONE]

> *The manuscript would also benefit from additional controls or comparative analyses. At present, there is no clear baseline against which the proposed framework can be evaluated. Possible additions include comparison with existing spatial transcriptomics or ligand-receptor methods, simulated datasets with known ground truth, spatial shuffling controls, ligand-receptor pair shuffling controls, random ligand-receptor combinations, or comparison with previously annotated tissue regions. These analyses would help determine whether the method identifies biologically meaningful spatial structure beyond smoothing, increased feature dimensionality, or clustering.*

We judged this to be the most important comment in the review, and we have addressed it in two ways.

The first concerns the reviewer's request for a baseline against which the framework can be evaluated. In the revised manuscript every comparison of interaction fields, between regions, along transects and over whole fields, is set beside three baselines: the plug-in map read directly, a Wilcoxon test that treats each location as an independent observation, and our own posterior with the correlation between locations removed (Methods; Results, Table 2). This shows where the framework changes a conclusion, and it separates the effect of ignoring spatial correlation from the effect of ignoring uncertainty altogether: across 19,201 region comparisons the three find differences in 90%, 73% and 46% of cases for the Wilcoxon test, the decorrelated posterior and the joint posterior respectively.

The second concerns the tissue domains, and is the controls from the reviewer's menu: a spatial permutation test with full model refitting, a comparison with previously annotated tissue regions, and ligand-receptor pair shuffling. They are summarized in the Results section on the domains and given in full in the new S16 Appendix; the code and summary outputs are in the public repository.

Throughout, spatial coherence of the resulting domain map is quantified with the entropy-based local indicator of spatial association (ELSA) computed on the categorical domain labels, at neighborhood radii of 50 and 100 units. ELSA is negatively oriented: lower values mean more spatially coherent domains. The domain-generating pipeline is held fixed across observed and control runs, so any difference is attributable to the manipulation alone.

Control A: coordinate shuffle with full model refitting

This is the direct test of whether the spatial smoother manufactures coherent structure from data with no spatial signal. We permuted the assignment of observed counts to spatial locations, destroying spatial structure while preserving the count distribution exactly, and then refit every feature's spatial model from scratch on the permuted data before running the identical convolution, clustering, and scoring pipeline. This required refitting all 912 features under each of 49 permutations, which we carried out on a high-throughput computing cluster. Because the clustering then had to be repeated for every permutation, both the observed and the permuted data were clustered with a faster implementation of the same steps rather than with Seurat; the pipeline is identical between the two arms of the control, which is what the comparison requires, and the manuscript now states this.

The result is unambiguous. The real data yielded 20 domains with ELSA at radius 50 of 0.075. The 49 coordinate-shuffled replicates yielded 4 to 12 domains with ELSA at radius 50 averaging 0.470, ranging from 0.412 to 0.583. Not one of the 49 shuffles came within a factor of five of the observed coherence; the null mean is 6.3 times the observed value. The permutation p-value is 0.020, which is the smallest value attainable with 49 permutations, so the test is limited by the number of replicates rather than by the separation between the observed value and the null.

We regard this as the substantive answer to the reviewer's concern. Fitting a spatial Gaussian process to spatially meaningless data does not produce coherent tissue domains; it produces a small number of incoherent fragments. The domains we report therefore reflect spatial structure present in the data, not structure imposed by the smoother.

Control B: concordance with independent anatomical annotation

As a positive control we compared the emergent domains to the anatomical annotations published with the source dataset by Chen et al. (2022) for the same tissue section. These annotations were produced by a different group, by a different method, and without reference to our labels, which our pipeline in turn never sees, so they are an independent reference in the sense that matters for a positive control.

For transparency we also checked and now state how those labels were derived: by spatially constrained clustering, combining a spatial nearest-neighbor graph with an expression nearest-neighbor graph, followed by manual labeling of each cluster from its marker genes against a reference embryo atlas. The comparison is thus between two partitions of the same tissue section, which is unavoidable, since a concordance check is only meaningful against annotations of the same specimen. The two partitions are nonetheless built from very different features, theirs from binned expression across all genes and ours from the inferred interaction fields of 1477 ligand-receptor pairs, and the manual labeling step brings in anatomical knowledge from outside the dataset. We note in the manuscript that the two controls answer different questions: the coordinate shuffle asks whether the structure is real, and the concordance comparison asks whether the structure that survives is anatomically meaningful.

All 26,000 modeled spatial bins matched an annotated bin. Agreement was substantially above chance: adjusted Rand index 0.401 and normalized mutual information 0.604, against a permutation null whose mean adjusted Rand index was 0.00003 and whose maximum over 499 permutations was 0.001, giving $p = 0.002$. The mean size-weighted purity of the emergent domains with respect to anatomical labels was 77.6%. Individual domains recover recognized structures cleanly: spinal cord at 97% purity, liver at 96%, one heart domain at 99%, meninges at 92%, with lung, muscle, thymus, and cartilage primordium also recovered.

Again we state the result carefully. An adjusted Rand index of 0.401 is substantial but far from perfect agreement, and the disagreement is structured rather than random: our signaling domains subdivide several organs, most visibly the heart, into multiple territories. We interpret this as the expected consequence of measuring a different modality. Anatomical annotation partitions tissue by cell-type composition, whereas our domains partition it by ligand-receptor signaling context, and the two need not coincide. The appropriate conclusion is that the emergent domains recover major anatomical structures while resolving additional substructure, not that they reproduce the anatomical segmentation.

Control C: ligand-receptor pair shuffling

We also ran the pair-shuffling control the reviewer suggests. Receptors were reassigned to ligands at random, and any reassigned pair that coincided with a real FANTOM5 pairing, or paired a gene with itself, was then repaired by targeted swaps, so that no shuffled set retained a real pair. The set of receptors is unchanged, so every shuffled set is built from exactly the same fitted fields as the true pairing, and only the pairing differs. Each shuffled set then passed through the same convolution, clustering and scoring as the true pairing.

Our first attempt at this control could not separate signal from noise: it used the faster clustering implementation described below in a configuration that did not reproduce exactly between runs, and the difference between the true pairing and the shuffled null was smaller than the run-to-run spread. We have since rerun it, in that implementation with a single thread and a fixed seed, which makes it exactly reproducible, and separately in Seurat, the implementation behind the original Figures 5 and 6 (now S15 Fig and Fig 9).

The true pairing did not produce more spatially coherent domains than shuffled pairings. In Seurat, with 19 shuffled sets, ELSA at radius 50 was 0.124 for the true pairing against a shuffled mean of 0.093 (standard deviation 0.018), and 17 of the 19 shuffles were more coherent than the true pairing. In the faster implementation, with 99 shuffled sets, the true pairing sat inside the shuffled distribution, with 24 of the 99 shuffles more coherent. The direction of the small difference is not consistent between implementations, and we read the result as no detectable effect of the pairing on domain coherence.

That is not the same as the pairing carrying no information. True pairings produced 1449 numerically distinct interaction fields. This count needs no clustering, so we computed it for 9,999 shuffled pairings: they produced 1424 on average (standard deviation 5.9), and only one reached 1449 or more, a permutation p of 0.0002. True partners are also more alike in spatial pattern than random partners matched for how consistently one partner is the smaller of the two (mean Pearson correlation 0.111 against 0.068 across the 1477 pairs; paired t-test $p = 4 \times 10^{-21}$), although even random partners are positively correlated, so the pairing-specific part is modest. Our working interpretation, which we have not tested directly, is that the common-minimum operator on raw counts passes little of this information on to the clustering: for 42.6% of pairs the same partner is the smaller one at more than 95% of locations, so the interaction field is largely that partner's field.

We have reported this control in the Results section on the domains and in S16 Appendix, and added a paragraph to the Discussion stating the limit it places on our conclusions: the domains reflect the collection of fitted ligand and receptor fields, not the specific pairings. We think this is the most useful thing the control tells a reader, and we would rather state it than leave the reviewer to infer it.

Reproducibility of the clustering, and which implementation was used where

Two implementations of the clustering step appear in this work, and we have made explicit in the manuscript which was used where and how reproducible each is. The original Figures 5 and 6 (now S15 Fig and Fig 9) were clustered with Seurat. Control A, and one arm of Control C, were clustered with a faster implementation of the same two steps (an RcppHNSW nearest-neighbor graph followed by Louvain clustering in igraph), because Control A needed the clustering repeated for every permuted dataset. The two give ELSA values on different scales, so each control compares observed and null within one implementation, and the manuscript now says which was used where.

The faster implementation is exactly reproducible with a single thread and a fixed seed, and not otherwise: across eight replicate runs on identical input with four threads and no seed, ELSA at radius 50 ranged from 0.053 to 0.067. For Seurat we have not established exact reproducibility. As an indirect check, an independently written script re-ran the published feature-depth sweep at six depths and at full depth: mean ELSA at radius 50 was within 0.012 of the published value at every depth, and 0.1236 against the published 0.1237 at full depth, with 19 domains against the published 20. The manuscript now treats each Seurat domain map as a single realization with an

approximate domain count.

What the controls establish together

Control A shows that the spatial structure is real rather than an artifact of smoothing. Control B shows that the resulting structure corresponds to recognized anatomy, while also resolving substructure within it. Control C shows that this structure does not depend on the specific ligand-receptor pairings, which narrows what it says about signaling. Together we believe these address the reviewer's question of whether the method finds meaningful structure beyond smoothing, dimensionality, and clustering: the coordinate shuffle rules out the smoothing and dimensionality explanations, since the shuffled data pass through an identical pipeline of identical dimensionality, and the anatomical concordance shows the surviving structure is biologically interpretable.

We have not added simulated datasets with known ground truth, the remaining item on the reviewer's menu. Our reasoning is that a simulation would have to be generated under a spatial model, and any simulation we could write would be close enough to our own fitted model that recovering the truth would demonstrate self-consistency rather than validity. The coordinate-shuffle control tests the same concern using the real data and without that circularity.

R2.9. Reproducibility documentation

[DONE]

> *Reproducibility also needs improvement. Although the repository contains a substantial amount of code, the execution order of scripts, required packages and versions, required input files, intermediate objects, figure-generating scripts, and complete workflow are not sufficiently documented. An external researcher would likely need to reverse engineer the analysis to reproduce the main results.*

This criticism was fair and we have acted on it. The repository now documents, for each analysis, the execution order of the scripts, the required input files and where to obtain the public ones, the intermediate objects each stage produces and consumes, and the package versions used.

We have also drawn an explicit line between artifacts that are tracked in the repository and large derived objects that are not. Every untracked derived object is now regenerated by a tracked script, and the documentation names that script. Where a generating script did not survive, we say so explicitly rather than implying the artifact is reproducible.

Concretely, the main pipeline under `Supplements for Publication/code/` now carries a document at its root giving the eight analysis stages in execution order, and for each one the script that runs it, the inputs it needs and where the public ones are obtained, the intermediate objects it produces and consumes, and the parameters that were set. Two sibling documents cover the two code directories in detail: the model specification, the priors, the mesh construction, and the convolution operators for the modeling scripts, and the clustering configurations and thresholds for the analysis scripts. A separate table states which script generated each manuscript figure, including the two figures whose composite assembly was done by hand and for which no layout script exists; we say that plainly rather than implying those composites are scripted. Package versions are given, with a full `sessionInfo()` captured under `code/environment/`. The repository root README now points a reader to these rather than leaving them to be found.

We have also documented, rather than quietly repaired, the places where the archived code does not run as written, and the places where a fresh run would legitimately differ from the archived outputs. Two defects that prevented the archived modeling scripts from running at all were transcription

errors introduced when the code was packaged, a stale relative path and an undefined flag; those we did fix, and the fix is recorded in the same document so that it is visible and reversible.

We should be clear about one limit that documentation cannot remove. The package environment of the original 2025 model fits was not recorded at the time, the fitted model objects were too large to retain, and the prediction objects carry no version metadata. The environment we supply is therefore a working one, verified to run the pipeline and used for the controls in Section R2.8, not the one that produced the archived fits. Table 1 reproduces exactly from the archived per-feature score files; we do not claim that a fresh refit under a different INLA version would reproduce it to the last digit, and we say so in the documentation rather than leaving a reader to discover it.

The new analysis is documented to the same standard, in `code-spatial-smoothing/uncertainty/` of the repository. It contains the design that fixed the regions, transects, thresholds and examples before any comparison was computed, with a dated log of every later addition; the row-by-row comparison of our refits with the published fitting procedure; the scripts for the local pilot, for the refits and per-pair computations on the computing cluster (with the measured memory and run time of each stage), and for the tabulation; and the scripts that draw Figs 5 to 8 and build S11 to S14. The posterior draws are large (about 41 GB for all genes) and are not stored, but every quantity in the manuscript is computed from per-pair summaries that are (S14 File), and the draws can be regenerated from the tracked scripts.

R2.10. Scientific positioning relative to existing work

[DONE]

> *The scientific positioning of the manuscript should also be strengthened. The authors should more clearly discuss related work in Gaussian Process-based spatial transcriptomics, Bayesian spatial count modeling, spatial variance component analysis, and spatial ligand-receptor inference. Relevant methods such as SpatialDE, BOOST-GP, Spatial Variance Component Analysis, and SpatialDM should be discussed to clarify which aspects of the manuscript are novel and which represent combinations, extensions, or reinterpretations of existing approaches.*

We have added a positioning subsection that discusses each of these method families and states plainly which parts of our contribution are new and which are combinations or reinterpretations of established ideas.

In outline: SpatialDE (Svensson et al., *Nature Methods*, 2018) introduced Gaussian-process testing for spatially variable genes, decomposing each gene's expression variance into a kernel-parameterized spatial component and a noise component and testing the former by a likelihood ratio test. We share its use of a GP prior over space but not its goal, since we estimate and use the field itself rather than testing features for spatial variability. BOOST-GP (Li et al., *Bioinformatics*, 2021) is the closest methodological relative: it places a gene-specific zero-mean Gaussian process on the spatial random effect inside a zero-inflated negative binomial count model, with Bayesian variable selection over whether that spatial component is present, and its stated goal is likewise identifying spatially variable genes. Our contribution relative to it is not the count-GP combination, which we explicitly do not claim as novel, but the downstream use of the posterior fields as continuous objects that are convolved and then analyzed jointly. Spatial Variance Component Analysis (Arnol et al., *Cell Reports*, 2019) decomposes expression variance into intrinsic, environmental, cell-cell interaction, and noise components and tests whether the interaction component is nonzero, which is a variance-partitioning aim rather than a field-estimation one, and it was developed for imaging-based data. SpatialDM (Li et al., *Nature Communications*, 2023) tests ligand-receptor pairs for

spatial co-expression using a bivariate extension of Moran's I, globally and per spot, with significance from a permutation null or an analytically derived one; the analytical null assumes that each gene is spatially randomly distributed. It targets significance of individual interactions where we target a continuous inferred field per interaction with quantified uncertainty.

We should correct one thing we had wrong in our own preparation, since it affects how we cite BOOST-GP. Its published title is "Bayesian modeling of spatial molecular profiling data via Gaussian process" and it appeared in *Bioinformatics*; "BOOST-GP" is the software name and does not appear in the journal title. We verified the characterization of each of these methods against the papers themselves rather than from summary knowledge, as the reviewer's comment warrants.

We have also added a discussion of CytoSignal (Liu et al., *Nature Genetics*, 2026), concurrent work that appeared after our submission and that analyzes the same Stereo-seq mouse embryo dataset. It infers spatially resolved ligand-receptor signaling at cellular resolution using a score with a spatial-permutation null, and validates against experimental proximity-ligation ground truth. We do not claim novelty for spatially resolved ligand-receptor mapping in light of that work, and we have removed language that implied it.

Reading these methods closely also clarified where our contribution lies, and it led to the reframing described at the start of this letter. None of the four methods the reviewer names, nor CytoSignal, uses a posterior over continuous fields to judge whether a difference visible on a map is supported by the data. We therefore read the closest precedents for that question in full and now discuss them as well: Splotch (Äijö et al., 2019), a Bayesian count model that compares anatomical regions one pair at a time through region coefficients; the excursion sets of Bolin and Lindgren (2015, 2018), applied with the same INLA and SPDE machinery to functional brain imaging by Mejia et al. (2020); SHADE (Eliason et al., 2026), which builds simultaneous credible bands from joint posterior draws for interaction curves between cell types; SpaceBF (Seal and Neelon, 2026), which places pointwise credible intervals on ligand-receptor co-expression; and simulation studies reporting that tests treating locations as independent overstate evidence under spatial correlation (Vasconcelos et al., 2026; Seal and Neelon, 2026).

Stating our contribution precisely, and more narrowly than the original submission did: we contribute a generative Bayesian geostatistical treatment of raw spatialomics counts that yields a posterior for every gene field and, through joint posterior draws, for every ligand-receptor interaction field; the use of those posteriors, with joint inference, to ask whether differences visible on interaction-field maps are supported, for regional averages, along transects and over whole fields, with the regions and examples fixed in advance; and the finding that clustering the full collection of inferred interaction fields produces tissue domains that correspond to anatomy (though, as the pair-shuffling control shows, not domains that depend on the specific ligand-receptor pairings). None of the ingredients is new on its own; the combination, applied to ligand-receptor interaction fields, is what we offer, and we now say so.

We have also corrected an overstatement of our own that this comparison brought into focus. Earlier drafts described the uncertainty as propagated through to the interaction fields. In the original analysis it was not: the convolution and every stage downstream of it operated on posterior medians, so the interaction fields were point estimates. The revised analysis does propagate it, for the comparisons. We refitted every gene with the published model to obtain joint posterior draws, checked that the refits reproduce the published fits, and formed the posterior of each interaction field draw by draw. The revised manuscript also reports how the plug-in fields differ from their posteriors (Fig 5): relative to the posterior median of the interaction, the plug-in field is never lower,

and is higher by more than 25% at 22% of pair-bin combinations, most where neither partner is clearly the limiting one, while the spatial pattern changes little (median correlation over bins 0.988). Carrying the posterior through the clustering itself remains undone, and the Discussion says so.

All of this now appears in a new Discussion subsection, "Relationship to existing methods," and all of these works are cited. That subsection also states plainly that we have not run a head-to-head benchmark against any of them, and why: there is no shared target quantity on which an accuracy comparison would be meaningful without first defining a benchmark task and a ground truth, which we regard as a separate piece of work. We would rather say that than leave the omission implicit.

R2.11. Generalizability claims

[DONE]

> *Some claims of generalizability appear broader than the evidence provided. The study is demonstrated using a selected region from a public Stereo-seq mouse embryogenesis dataset. This is useful as a proof of concept, but spatial omics platforms differ in resolution, capture efficiency, sparsity, noise structure, segmentation assumptions, field of view, and biological scale. The authors should moderate broad claims about applicability to most spatial omics datasets or provide additional analyses and practical guidance for different platforms.*

We accept this and have moderated the claims throughout, rather than attempting to justify the original ones with additional platforms.

We should, however, separate two things that the original text ran together. One is whether the framework is in principle tied to Stereo-seq, and we do not think it is: the model needs only counts at known locations with a measure of local sampling effort, and the SPDE representation of the Gaussian process is built on a triangulated mesh rather than a lattice, so irregular or hexagonal observation geometries need no special handling. We have separately fitted fields of this kind to 10x Visium mouse brain sections, a hexagonal spot array at much coarser resolution, for 2,035 ligand-receptor mechanisms across two sagittal sections, with posterior predictive occurrence tracking observed occurrence closely across all 3,865 fits (correlation 0.99, median absolute difference 0.015). The other thing is whether the *pipeline* has been shown to transfer, and it has not. That earlier work used a two-part semicontinuous observation model on normalized data rather than the raw-count model developed here, and it stopped at per-mechanism fields without going through the convolution and clustering stages. We have added a short passage to the Discussion making exactly this distinction, and stating that comprehensive evaluation across tissues and platforms remains future work. We would rather record the limited evidence we have with its limits attached than either overclaim from it or leave the reader with the impression that the method is inherently platform-specific.

The revised manuscript states the scope of the demonstration explicitly wherever a general claim previously appeared: one field of view, from one section, of one embryo, on one platform, at one binning resolution. Claims of the form "applicable to spatial omics data" have been narrowed to describe what we demonstrate, with the extension to other platforms stated as an expectation and its conditions given rather than asserted.

We have added a short paragraph on what would need to hold for the approach to transfer, which we think is more useful to a reader than a broad claim: the observations must be counts with an interpretable exposure or offset, the spatial support must be dense enough that a Gaussian process is identifiable at the length scales of interest, and at single-cell-resolution platforms the binning step we rely on would need to be replaced by an explicit treatment of cell geometry. We also note

that platforms with substantially sparser capture may favor a zero-inflated likelihood, which our Table 1 comparison found unnecessary at this sparsity but which the framework accommodates.

The specific changes are as follows. The Abstract previously asserted that "the general methods piloted here can readily be applied to spatialomics data from diverse platforms with no need to alter data collection techniques"; it has been rewritten to state what was actually done and on what, and it now ends by naming the scope of the demonstration and pointing to the transfer conditions instead of asserting general applicability. The opening sentence of the model section previously claimed the model "can be applied to most, if not all, existing spatialomics datasets"; it now states the property a dataset must have for the model to apply, and cross-references the Discussion. The Discussion claim that "our methods are generally-applicable to any spatial transcriptomic data" has been replaced by the paragraph of transfer conditions described above.

One further sentence needed correcting rather than softening. The limitations paragraph said that "our general approach has thus far proved easy to adapt on all data types tested," which is not supportable in a manuscript that fits only Stereo-seq data. It now says that we have fitted the model only to Stereo-seq data here, so our expectation that adaptation is straightforward is an expectation rather than a demonstrated result.

R2.12. Caution in biological interpretation

[DONE]

> *Finally, the biological interpretation should remain cautious. I appreciate that the authors acknowledge that transcript abundance is only used as a proxy for protein abundance. However, because downstream analyses involve inferred ligand-receptor interactions and morphogenic signaling fields, the manuscript should avoid language that implies direct evidence of protein-level interactions, ligand-receptor binding, signaling activity, or information transmission. Terms such as "proxy," "inferred field," "model-derived estimate," or "potential interaction" would be more appropriate.*

We agree, and we have revised the language throughout.

We have replaced formulations that implied observed protein-level events with formulations describing model-derived estimates from transcript measurements. Descriptions of ligand-receptor quantities now consistently use "inferred interaction field" or "potential interaction" rather than wording that implies binding was observed; references to signaling activity are now phrased as transcriptional evidence consistent with potential signaling; and language implying information transmission between regions has been removed or explicitly marked as interpretation.

We have also added an explicit statement of the inferential chain and its assumptions in the Discussion, so a reader can see exactly how far the claims are from the measurement: transcript counts are a proxy for transcript abundance, transcript abundance is a proxy for protein abundance, co-localization of ligand and receptor proxies is a proxy for the opportunity for interaction, and none of these steps is directly observed here. We now also state the scope of the method plainly, in the Abstract and the Discussion: the interaction fields estimate the potential for signaling under the co-expression assumption standard in cell-cell communication analysis, and we propose them as a proxy for signaling rather than a measurement of it. The Discussion also notes that co-presence is assessed within a bin of 50 by 50 capture spots rather than at a shared cell surface.

The inferential chain now appears explicitly in the Discussion, in the terms the reviewer suggests: transcript counts are a proxy for local transcript abundance; transcript abundance is a proxy for protein abundance, a step we do not attempt to correct; co-localization of a ligand proxy and a

receptor proxy is a proxy for the *opportunity* for interaction; and none of the intermediate quantities, and no binding event, is directly observed in this work.

We settled on "inferred interaction field" as the single term for the convolved output and use it consistently. Specific replacements include: a claim that the model outputs "directly estimate local ligand-receptor binding density" now says they estimate a proxy for it; "the maximum possible number of local binding events" is now an upper bound implied by the estimated levels under a stated assumption; "individual ligand-receptor binding density field" in the discussion of the domain figure (now Fig 9) is now "individual inferred ligand-receptor interaction fields"; and the invitation to readers to "observe the true scale and structure of extracellular communication in mammalian tissues" now invites them to explore the scale and structure of the inferred fields. We also removed several intensifiers that editorialized rather than described, including "statistically-powerful," "experimentally pioneered," and two uses of "significant" in a non-statistical sense.

7. Additional corrections

[DONE]

Internal note: the item below was found by us, not by a reviewer. It should still be disclosed in the letter, since the reviewer will notice the numbers changed.

In preparing this revision we audited the feature and interaction counts reported in the original manuscript against the analysis outputs, and found errors that we have corrected throughout. We report them here because they change numbers and statements that appear in the text, in Table 1, and in the original Figures 5 and 6 (now S15 Fig and Fig 9).

First, the attrition from the curated feature list was not described. The list contains 918 ligand or receptor genes. Three (*a*, *Calcb*, *Cckar*) were excluded before fitting, and three more (*Fgf1*, *Fzd7*, *Trhr*) failed to converge, leaving 912 features with fitted spatial fields. The original text stated that the spatial fields of all 918 features were modeled. The revised text gives the correct figure and explains the attrition.

Second, the number of interaction fields was overstated by four. The manuscript reports 1481 unique ligand-receptor interaction fields, and describes them as following from the 918-gene list. Neither part of that is right. The 1481-row object used for the high-dimensional clustering contains four exactly duplicated rows, which we traced to files that had been duplicated in the input files and then read in alongside their originals. The four affected interactions are PDGFC-PDGFRA, PDGFC-PDGFRB, PDGFD-PDGFRA, and PDGFD-PDGFRB, and in each case the duplicate is numerically identical to its counterpart. The correct number of ligand-receptor interactions computable from the 912 fitted features is therefore 1477, which we confirmed by re-deriving it from the FANTOM5 pairings and checking it against the 1481 rows of the clustered object.

These 1477 interactions are not 1477 numerically distinct fields. Because each field is the pointwise minimum of its ligand and receptor fields, interactions that share a partner which is the smaller of the two at every location all reduce to that partner's field. For example, seven interactions of the ligand GDF9 with different receptors yield one and the same field. Counting this way, the 1477 interactions give 1449 numerically distinct fields. This is a property of the minimum operator rather than an error, and we now state it in the manuscript alongside the count of interactions. Separately, 1481 was never the right figure for the 918-gene list either: 1493 FANTOM5 pairs have

both partners among the 918 curated genes. We have corrected the count wherever it appears, including the upper limit of the feature-depth sweep in the domain figure (now Fig 9).

We do not expect this to affect the results materially, but we have not measured it directly. The duplication gives four of the 1477 interactions twice their intended weight in a clustering over more than a thousand scaled dimensions. As an indirect check, a re-run of the full-depth clustering without the duplicated columns, by an independently written script, gave a mean ELSA at radius 50 of 0.1236 against the published 0.1237, and 19 domains against 20. Because we have not established that the Seurat clustering is exactly reproducible (see R2.8), that comparison bounds the combined effect of the duplication and of run-to-run variation rather than isolating the duplication. The manuscript now discloses the duplication in the Materials and methods.

Both corrections have been applied to every occurrence in the manuscript text, in the Table 1 caption, and in the domain figure caption, and to the panel labels inside the domain figure image, which previously read "1481 Features".

Re-deriving the feature-depth experiment (now Fig 9) from its saved outputs also showed that three statements about it were wrong, and we have corrected them.

The caption said that domains become more numerous as features are added. They become fewer: 33 at 10 features, 24 at 150, and between 16 and 21 from 350 onward. The Results said that the full feature set gave the greatest spatial coherence of the depths swept. Five deeper depths (1200 to 1450 features) scored slightly better, although differences of that size are close to what a re-run of the clustering can produce. The Results also said that clustering without principal-component analysis gave the greatest complexity and coherence of the configurations swept. We found no matched comparison with and without principal-component analysis in our analysis record, so we have removed that claim and now say only that the clustering was run without it.

Because the published experiment admitted features in order of variance, it confounded the number of features with which features they were. We have added a control that admits random subsets of the same sizes (19 per depth). Mean ELSA at radius 50 then falls steadily with depth, from 0.372 at 10 features to 0.120 at 1000, with most of the change by a few hundred features. The rise and fall of coherence in the published curve does not appear with random subsets, which suggests it reflects the order of admission rather than the number of features. Both results are now in the Results.

Finally, the operator-comparison caption (now S15 Fig) and text said that the common minimum gave the most clusters at the lowest spatial entropy of the operators tested. It gave the most clusters, but the geometric mean had slightly lower spatial entropy (ELSA 0.047 against 0.050 at radius 100), a difference within run-to-run variation. The text now says so, and no longer ranks those two operators on coherence.

One further finding is disclosed for completeness. To obtain the joint posterior draws used in the revised analysis we refitted all 790 genes needed by the ligand-receptor pairs with the published specification, using current versions of the fitting software, since the versions used for the published fits were not recorded. For 778 genes the refits reproduce the published per-location posterior medians and interval widths within Monte Carlo error. For 12 sparsely detected genes they do not, and we could not establish why. Those genes enter 17 of the 1,477 pairs; we flag these pairs rather than exclude them, and every count in the revised manuscript is also reported without them, changing none by more than about 1% (S13 Table).